

# A Practical Guide to Hierarchical clustering

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TOPIC -- HIERARCHICAL CLUSTERING

-- 01 --

$$d(\mathbf{x}, \mathbf{y}) = \frac{\|\mathbf{x} - \mathbf{y}\|}{S^{1/\top}} \quad \{\displaystyle d(\mathbf{x}, \mathbf{y}) = \frac{\|\mathbf{x} - \mathbf{y}\|}{S^{1/\top}}\}$$

-- 02 --

In data mining and statistics, hierarchical clustering (also called hierarchical cluster analysis or HCA) is a method of cluster analysis that seeks to build a hierarchy of clusters. Strategies for hierarchical clustering generally fall into two categories:

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**Cluster linkage** In order to decide which clusters should be combined (for agglomerative), or where a cluster should be split (for divisive), a measure of dissimilarity between sets of observations is required. In most methods of hierarchical clustering, this is achieved by use of an appropriate distance  $d$ , such as the Euclidean distance, between single observations of the data set, and a linkage criterion, which specifies the dissimilarity of sets as a function of the pairwise distances of observations in the sets. The choice of metric as well as linkage can have a major impact on the result of the clustering, where the lower level metric determines which objects are most similar, whereas the linkage criterion influences the shape of the clusters. For example, complete-linkage tends to produce more spherical clusters than single-linkage. The linkage criterion determines the distance between sets of observations as a function of the pairwise distances between observations. Some commonly used linkage criteria between two sets of observations  $A$  and  $B$  and a distance  $d$  are:

-- 04 --

**Divisive clustering** The basic principle of divisive clustering was published as the DIANA (Divisive ANALysis clustering) algorithm. Initially, all data is in the same cluster, and the largest cluster is split until every object is separate. Because there exist  $O(2^n)$  ways of splitting each cluster, heuristics are needed. DIANA chooses the object with the maximum average dissimilarity and then moves all objects to this cluster that are more similar to the new cluster than to the remainder. Informally, DIANA is not so much a process of "dividing" as it is of "hollowing out": each iteration, an existing cluster (e.g. the

initial cluster of the entire dataset) is chosen to form a new cluster inside of it. Objects progressively move to this nested cluster, and hollow out the existing cluster. Eventually, all that's left inside a cluster is nested clusters that grew there, without it owning any loose objects by itself. Formally, DIANA operates in the following steps:

-- 05 --

Because it disregards vector magnitude, it is frequently used in text mining and document clustering, where documents are represented as high-dimensional term-frequency vectors. Mahalanobis distance accounts for correlations between features by incorporating the inverse of the data's covariance matrix  $S$ :

-- 06 --

Selecting the number of clusters In hierarchical clustering, the number of clusters is not determined automatically and must be chosen by analyzing the dendrogram. A common approach is to “cut” the dendrogram at a selected height, which partitions the data into clusters based on where the cut intersects the tree structure. The choice of cut level has a significant impact on the resulting clustering. One widely used heuristic is to identify large vertical gaps in the dendrogram, which correspond to substantial increases in the distance between successive merges. Cutting the dendrogram at a height just before such a gap can produce clusters that reflect natural groupings in the data. This is based on the observation that merges occurring at much larger distances often combine dissimilar clusters.

-- 07 --

Distance metrics While the linkage criterion determines how dissimilarity between sets of observations is computed, the underlying distance metric determines how dissimilarity between individual observations is measured. Because hierarchical clustering permits any valid measure of distance, the choice of metric is guided by the nature of the data and can have a significant effect on the resulting clustering. Euclidean distance is the most widely used metric for continuous numerical data. It corresponds to the straight-line distance between two points in Euclidean space and is the default choice in most statistical software:

-- 08 --

It is often preferred when features are measured on different scales or when the data contain outliers, as it is less sensitive to large deviations than Euclidean distance. Cosine distance measures angular dissimilarity between two non-zero

vectors:

-- 09 --

$$\frac{1}{|A| \cdot |B|} \sum_{x \in A} \sum_{y \in B} d(x, y) \quad \{\displaystyle \frac{1}{|\mathcal{A}| \cdot |\mathcal{B}|} \sum_{x \in \mathcal{A}} \sum_{y \in \mathcal{B}} d(x, y) \}$$

-- 10 --

Interpreting dendrogram height A key aspect of interpreting a dendrogram is understanding the meaning of branch height. The height at which two clusters are joined represents the distance or dissimilarity between those clusters, where lower merges indicate greater similarity and higher merges indicate greater dissimilarity. Early merges at lower heights reflect the closest relationships between observations, while later merges combine increasingly different groups. When the dendrogram is scaled using actual or normalized distances, the vertical spacing between merges provides insight into the relative differences between clusters. Large vertical gaps between successive merges often indicate meaningful separations in the data and can help guide the selection of cluster boundaries, although dendrograms alone should not be used to definitively determine the number of clusters. Because dendrograms summarize underlying distance relationships, some information may be lost in the visualization, and interpretations are generally most reliable for clusters that merge at lower heights.

-- 11 --

This definition captures the largest distance from a point in one set to the closest point in the other set. Equivalently, it can be interpreted as the smallest radius  $r$   $\{\displaystyle r\}$  such that every point of  $A$   $\{\displaystyle A\}$  lies within distance  $r$   $\{\displaystyle r\}$  of some point in  $B$   $\{\displaystyle B\}$ , and every point of  $B$   $\{\displaystyle B\}$  lies within distance  $r$   $\{\displaystyle r\}$  of some point in  $A$   $\{\displaystyle A\}$ . In this sense, the Hausdorff distance measures the maximum mismatch between two sets. In hierarchical clustering, Hausdorff linkage uses  $d_H(A, B)$   $\{\displaystyle d_H(A, B)\}$  as the dissimilarity measure between clusters. Initially, each data point forms its own cluster, and the Hausdorff distance between singleton sets reduces to the original metric. At each step, the two clusters with the smallest Hausdorff distance are merged, and distances are recomputed using the same set-based definition. This iterative process continues until all points belong to a single cluster, producing a hierarchical structure (dendrogram). Compared to other linkage criteria such as single, complete, or average linkage, Hausdorff linkage operates on a set-to-set basis

rather than relying solely on pairwise point distances. As a result, it captures the global geometry of clusters and can better preserve structural features of the data. It has been shown to provide behavior intermediate between single and complete linkage, avoiding the chaining effect of the former while reducing the tendency of the latter to overestimate distances between clusters. However, these advantages come with increased computational cost. Computing the Hausdorff distance between two clusters typically requires evaluating distances between all pairs of points across the sets, often via a max–min procedure over a distance matrix. This makes Hausdorff linkage more computationally expensive than simpler linkage criteria, particularly for large datasets. Overall, Hausdorff linkage provides a mathematically well-founded clustering approach that is effective for distinguishing complex structures in data.